Feature Selection for Distribution Shift via Meta-Learner

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1 Aggregate the distribution shift variable importance through federated aggregation

In federated learning, model outputs are aggregated across sites in later stages, with the aim of preserving as much information as possible. This section asks whether FSDS, together with out-of-distribution variable importance, can sit in that pipeline: raw rows stay local, and each site uploads one score vector. The section answers four questions. When does the weight on a site change, and to what. How the aggregate responds when individual sites are corrupted. What the Matrix Bernstein argument actually supports. How a shift driver is justified when many sites are present and they do not all carry the same concept.

Federated aggregation keeps raw rows local and moves only scores or nuisance fits. For variable importance, [19] computes Shapley values for host features in vertical federated learning, while [6] constructs row-free SHAP backgrounds via federated clustering and GANs. [2] aggregates client-specific SHAP vectors using accuracy and faithfulness weights instead of a flat mean. For vertical feature selection, [7] ranks features via conditional mutual information on statistical summaries, and [10] refines this with Gini initialization and encrypted stochastic gates. To mitigate shift, [9] pools split statistics to grow a federated forest whose impurity decrease approximates a pooled forest under covariate and outcome shift. [8] builds global propensity scores from membership weights, yielding Fed-IPW and Fed-AIPW, and compares those plug-in estimators with meta-analysis. For causal effects, [21] ships propensity summaries to construct IPW and AIPW estimators, and [17] learns a posterior over potential outcomes via Gaussian processes across sites. [1] uses propensity-weighted FedAvg so that sites with scarce treated rows do not dominate the average.

FSDS is applied to the same task. Rank correlations are computed for MMD-LOCO, a random-forest domain classifier, and pseudo-outcome risk, each against a stated reference: centralized importance, the pooled wrapper, or whole-batch FSDS. These references are not interchangeable. Some information loss across isolated sites is unavoidable. For permutation variable importance and for PermuCATE, inverse-variance weighting accounts for uncertainty at each site. Sample-size weights and the coordinate-wise median are the other magnitude aggregators. The subsections below say which of these is the weight to use, using the robustness grids, the real-data splits, and the concentration bound.

1.1 When the weight on a site changes

Site k uploads one vector $V_k \in [0,1]^p$ with $\sum_j V_{k,j} = 1$. The aggregate is

$$\bar{V} = \sum_{k=1}^{K} w_k V_k, \qquad w_k \geq 0, \qquad \sum_k w_k = 1, \tag{1}$$

or the coordinate-wise median of the same uploads. The default is the uniform weight $w_k = 1/K$. It is the right default when the sites are noisy copies of one shift and are about the same size. The weight on site k changes only in the rows of Table 1. Communication remains one upload of p floats per site, $O(Kp)$ in total. No rule below asks for a second round.

Size is the first change. On the 245-cell grid of Table 4, with size in $\{0.1, \ldots, 0.9\}$, feature correlation in $\{0, 0.15, \ldots, 0.6\}$, and heterogeneity in $\{0, \ldots, 1\}$, the gain $\Delta(n|S| - \text{unif}) = \rho(n|S|) - \rho(\text{unif})$ is $+0.004$ in the low size-mismatch bin and $+0.285$ in the high bin. There the sites agree least with each other (pairwise concordance 0.105) and the size-weighted aggregates agree most with the mixed pool (0.905, against 0.620 for the uniform average). Rank aggregation moves the other way, from 0.658 to 0.580. Table 5 splits the same grid by the heterogeneity fraction. At high size mismatch the gain rises from $+0.24$ at hetero $= 0$ to $+0.37$ at hetero $= 1$, while $\rho(n|S|)$ rises from 0.893 to 0.927. Table 6 shows that this rise is present at every feature

Table 1: Per-site weight. ρ is Spearman correlation with the reference named in the row. Δ^{cf} is the site score (6).

What is observed	Weight on site k	Evidence
Sizes are close and the shift is shared	$w_k = 1/K$, or $n\|S\|$	TabReD max-frac = 0.24, $\Delta(n\|S\| -$ unif$) \approx 0.01$
Size mismatch is high	$w_k \propto n_k$, or $n\|S\|$	High bin: ρ from 0.620 to 0.905
One or two uploads are replaced	median, or client-agreement Shapley	Two random corruptions: median 0.741, uniform 0.055
Variance estimates are bounded below and the wrapper is a covariate score	$w_k \propto 1/\hat{\sigma}_k^2$	WhyShift domain classifier: 0.848 $\rightarrow$ 0.918
A corrupted site reports a small variance	floor $\hat{\sigma}_k^2$, or use the median	One bad site, variance ratio 0.44: IVW $\rho = 0.844$, mean $\rho = 0.959$
The wrapper is a concept score and sites need not share it	$w_k \propto \Delta_k^{\mathrm{cf}}$	PO-risk n-weight falls 0.410 $\rightarrow$ 0.185; Income shift AUC 0.869 $\rightarrow$ 0.917
Every weight returns the same ρ, and that ρ misses the whole batch	do not retune w_k	`num_105`: pair 0.884, $\rho \approx -0.20$, headroom 0

Table 2: Mixed synthetic blocks with corrupted client vectors. Spearman correlation with the centralized importance. Shapley is client agreement, not feature-wise Delta-Shapley.

Attack	# bad	Uniform	n-weighted	Median	Shapley
Random	1	0.208	0.539	0.798	0.802
Random	2	0.055	0.156	0.741	0.737
Extreme	1	0.503	0.576	0.830	0.814
Extreme	2	0.624	0.669	0.794	0.762
Collude	2	0.572	0.612	0.899	0.842

correlation from 0 to 0.60. Pairwise disagreement is the regime in which a size weight earns its keep, provided the large sites are the ones whose vector estimates the pool.

A replaced vector is not a small site. Table 2 replaces one or two client vectors. Two random replacements leave the uniform Spearman at 0.055 and the sample-size weight at 0.156. The median stays at 0.741 and client-agreement Shapley at 0.737. Client-agreement Shapley is a weight on sites. It is not Delta-Shapley, which is the feature-wise absolute difference of Shapley values between two batches and appears only in the single-batch comparison of Table 22.

Inverse-variance weights are the third change, and only under the variance bound in Assumption (A6) below. On ten random WhyShift folds (Table 3), the random-forest domain classifier moves from 0.848 on a single fold to 0.887 with sample-size weights and to 0.918 with inverse-variance weights. MMD-LOCO moves from 0.891 to 0.985 with sample-size weights. Both wrappers estimate a property of $P(X)$ shared across folds. Pseudo-outcome risk does not: a single fold correlates 0.410 with the pooled PO-risk vector, sample-size weights lower that to 0.185, and inverse-variance weights return it to 0.431. Averaging a concept vector as if it were another noisy copy of one concept makes the ranking worse. The concept row of Table 1 is the repair: set the weight proportional to the site score of Section 1.5, which is zero when the local model has no skill.

The last row is a refusal. Cooking-time `num_105` (Tables 19 and 20) has pairwise concordance 0.884 and site-to-whole correlation -0.198. Uniform, sample-size, $n|S|$, and rank all return a Spearman near -0.20. Under a feature-rank protocol the headroom $\Delta(n|S| - \text{unif})$ is $+0.01$; under a random array split the Spearman rises to about 0.80 and the headroom is 0. The split axis is the defect. Retuning w_k does not change it. Leave-one-site-out influence does not find it either (Section 1.3).

1.2 Robustness to corruption of individual sites

The pressure test uses many sites and asks whether the ranking of shifted features survives corrupted centers and a lower signal-to-noise ratio. The benchmark is a high-dimensional synthetic design rather than a small saturated table such as IHDP. Unless otherwise noted, local VIMP is an FSDS share, a random-forest importance, or $|\Delta\text{corr}|$. A dash means the variance ratio is undefined because one of the two groups is empty, or that the cell was not computed. The product $n_{\mathrm{bad}} \times \mathrm{SNR}_{\mathrm{low}} \times \alpha$ is Table 11, now with the inverse-variance

Table 3: FSDS on a WhyShift table, ten folds. Spearman correlation with the pooled out-of-distribution wrapper. The MMD-LOCO inverse-variance cell is not reported.

Wrapper output	Single fold	n-weighted	IVW
RF domain classifier	0.848	0.887	0.918
MMD-LOCO	0.891	0.985	—
PO-risk	0.410	0.185	0.431

Table 4: Simulation grid (245 cells; size $\in \{0.1, \ldots, 0.9\}$, correlation $\in \{0, 0.15, \ldots, 0.6\}$, hetero $\in \{0, \ldots, 1\}$): mean ρ by size-mismatch bin. Pairwise concordance is agreement among sites, not with the pool.

size bin	n	pair	unif	n-wt	$n\lvert S\rvert$	rank	$\Delta(n\lvert S\rvert - \text{unif})$
low (≤ 0.3)	70	0.241	0.651	0.660	0.655	0.658	+0.004
mid	105	0.191	0.658	0.786	0.784	0.647	+0.126
high (≥ 0.7)	70	0.105	0.620	0.905	0.905	0.580	+0.285

column included so that weight inflation can be read off the same grid. Entries are means over five replicates. w_{bad} is the weight mass on corrupted sites. It equals 0 at $n_{\text{bad}} = 0$ and 1 at $n_{\text{bad}} = 12$.

Selection and rank correlation break at different places. At full local support $\alpha = 1$ and SNR $= 0.5$, precision at K of the mean, the median, and inverse-variance weights is still 1 when every site is corrupted, while Spearman of the mean has fallen from 1 to 0.719. At SNR $= 0$ and twelve corrupted sites the selection AUC of the mean falls to 0.479 and its precision to 0. One remaining clean site is enough for the mean and not for the median: at $n_{\text{bad}} = 11$, SNR $= 0.05$, and $\alpha = 1$, precision of the mean is 0.92 and precision of the median is 0.14. Thin support hurts the median before any site is corrupted. At $\alpha = 0.25$, SNR $= 0.5$, and $n_{\text{bad}} = 0$, median precision is 0.29 and median AUC is 0.768, against precision 1 and AUC 1 for the mean and for inverse-variance weights.

Inverse-variance weights pay for a small reported variance. At one corrupted site, $\alpha = 1$, and SNR $= 0.5$, the variance ratio is 0.44 and $w_{\text{bad}} = 0.17$, about twice the equal share $1/12$. Spearman falls to 0.844 against 0.959 for the uniform mean, while precision stays 1. Assumption (A6) floors this variance so the weight stays bounded. The floor does not turn inverse-variance weighting into a detector of a bad concept. Table 10 records the five rows that carry this reading. Table 7 is the mean-share slice of the same grid at $\text{SNR}_{\text{low}} = 0.15$, read across local support $\alpha \in \{1, 0.75, 0.5, 0.25\}$. Table 8 records how far n_{bad} must rise before Spearman falls through 0.95, 0.90, and 0.80, and the SNR swap at a clean SNR of 1.5. Table 9 records the first corrupted-site count at which mean share and median share cross the same thresholds. The full product grid follows.

Table 5: Mean $\rho(n|S|)$, pairwise concordance, and $\Delta(n|S| - \text{unif})$ by size bin and heterogeneity fraction, averaged over feature correlation. At high size mismatch, Δ rises with hetero while pairwise falls.

	low size			mid			high size	
hetero	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	pair	Δ	$\rho_{n\|S}$	Δ
0	0.653	0.339	+0.01	0.768	0.297	+0.11	0.893	+0.24
1/6	0.639	0.283	+0.01	0.791	0.245	+0.12	0.886	+0.25
1/3	0.656	0.262	−0.00	0.769	0.215	+0.11	0.888	+0.27
1/2	0.640	0.250	+0.00	0.773	0.197	+0.13	0.892	+0.27
2/3	0.657	0.213	−0.00	0.793	0.145	+0.13	0.921	+0.29
5/6	0.664	0.183	+0.01	0.790	0.134	+0.12	0.925	+0.31
1	0.675	0.157	+0.01	0.804	0.103	+0.15	0.927	+0.37

Table 6: High size-mismatch cells only: $\Delta(n|S| - \text{unif})$ on the hetero × correlation grid.

hetero \ corr	0	0.15	0.30	0.45	0.60
0	+0.19	+0.23	+0.25	+0.27	+0.26
1/6	+0.25	+0.26	+0.27	+0.26	+0.23
1/3	+0.24	+0.27	+0.26	+0.33	+0.23
1/2	+0.22	+0.28	+0.26	+0.28	+0.31
2/3	+0.29	+0.28	+0.29	+0.31	+0.30
5/6	+0.29	+0.36	+0.30	+0.28	+0.28
1	+0.40	+0.36	+0.37	+0.35	+0.36

Table 7: High-dimensional robustness under varying local support α. Mean share at $\text{SNR}_{\text{low}} = 0.15$. Spearman ρ, selection AUC, and precision at K, as n_{bad} runs from 0 to 12.

	$\alpha = 1$			$\alpha = 0.75$			$\alpha = 0.5$			$\alpha = 0.25$		
$n_{\text{bad}}/12$	ρ	AUC	$P@K$	ρ	AUC	$P@K$	ρ	AUC	$P@K$	ρ	AUC	$P@K$
0	1.000	1.000	1.00	1.000	1.000	1.00	1.000	1.000	1.00	1.000	1.000	1.00
1	0.938	1.000	1.00	0.942	1.000	1.00	0.954	1.000	1.00	0.973	0.991	0.98
2	0.890	1.000	1.00	0.898	1.000	1.00	0.915	1.000	1.00	0.949	0.990	0.98
3	0.852	1.000	1.00	0.858	1.000	1.00	0.885	1.000	1.00	0.929	0.984	0.97
4	0.818	1.000	1.00	0.827	1.000	1.00	0.857	1.000	1.00	0.909	0.979	0.95
5	0.791	1.000	1.00	0.795	1.000	1.00	0.833	1.000	0.99	0.888	0.968	0.92
6	0.767	1.000	1.00	0.770	1.000	1.00	0.811	1.000	0.99	0.874	0.953	0.86
7	0.746	1.000	1.00	0.748	1.000	1.00	0.795	0.999	0.97	0.856	0.930	0.79
8	0.730	1.000	1.00	0.729	1.000	1.00	0.777	0.985	0.94	0.837	0.905	0.72
9	0.713	1.000	1.00	0.716	0.999	0.98	0.760	0.980	0.90	0.819	0.880	0.65
10	0.699	1.000	1.00	0.701	0.997	0.94	0.741	0.962	0.80	0.801	0.857	0.52
11	0.686	1.000	1.00	0.688	0.985	0.86	0.724	0.936	0.64	0.785	0.799	0.37
12	0.675	0.988	0.75	0.672	0.945	0.60	0.706	0.890	0.37	0.768	0.744	0.19

Table 8: Shared support: dimensional scaling and an SNR swap. The top panel is the smallest n_{bad} at which Spearman falls below the stated threshold. The bottom panel is AUC and precision at K when the clean SNR is 1.5.

		Tipping points (n_{bad})			$n_{\text{bad}} = 10$
p	K	$\rho < 0.95$	$\rho < 0.90$	$\rho < 0.80$	(ρ, AUC, $P@K$)
500	20	1	3	7	(0.718, 0.987, 0.81)
1000	25	1	2	4	(0.655, 0.981, 0.69)
2000	50	1	1	3	(0.538, 0.959, 0.57)

SNR swap at fixed p (clean SNR = 1.5, AUC = 1.0)

		$n_{\text{bad}} = 5$		$n_{\text{bad}} = 10$	
SNR_{low}	p	AUC	$P@K$	AUC	$P@K$
0.5	1000	1.000	1.00	1.000	1.00
0.5	2000	1.000	1.00	1.000	1.00
0.25	1000	1.000	1.00	1.000	1.00
0.25	2000	1.000	1.00	1.000	0.97
0.05	1000	1.000	1.00	0.628	0.10
0.05	2000	1.000	1.00	0.605	0.06
0	1000	1.000	1.00	0.487	0.02
0	2000	1.000	1.00	0.497	0.04

Table 9: Contamination tipping points for mean share and median share. Entries are the smallest n_{bad} at which the metric falls below the threshold. A dash means the threshold is not crossed on the reported grid.

		Mean share			Median share		
Mode	SNR_{low}	$\rho < 0.90$	AUC< 0.95	$J@K < 0.7$	$\rho < 0.90$	AUC< 0.95	$J@K < 0.7$
low SNR	0.5	—	—	—	—	—	—
low SNR	0.25	—	—	—	3	—	—
low SNR	0.1	10	10	10	3	9	9
low SNR	0	7	9	9	3	8	7
null SNR	—	5	9	9	3	8	7
wrong plant	—	1	7	6	4	6	6
y scramble	—	4	—	—	2	—	—

Table 10: Boundary rows of Table 11. ρ is Spearman with the planted ranking. AUC and P are selection AUC and precision at K.

n_{bad}	α	SNR	ρ mean	ρ IVW	P mean	P med	w_{bad}
0	0.25	0.5	1.000	1.000	1.00	0.29	0.00
1	1	0.5	0.959	0.844	1.00	1.00	0.17
11	1	0.05	0.677	0.651	0.92	0.14	0.97
12	1	0.5	0.719	0.710	1.00	1.00	1.00
12	1	0	0.606	0.601	0.00	0.02	1.00

Table 11: Product $n_{\text{bad}} \times \text{SNR}_{\text{low}} \times \alpha \times$ aggregator on the high-dimensional stress test. Entries are means over five replicates.

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
0	1	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.5	0.959	0.977	0.962	0.844	1.000	1.000	1.000	1.00	1.00	1.00	0.17	0.44
2	1	0.5	0.925	0.955	0.928	0.772	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.43
3	1	0.5	0.894	0.935	0.891	0.730	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.43
4	1	0.5	0.864	0.912	0.857	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.53	0.44
5	1	0.5	0.836	0.889	0.823	0.692	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	1	0.5	0.812	0.866	0.795	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	1	0.5	0.792	0.842	0.759	0.689	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	1	0.5	0.776	0.822	0.732	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	1	0.5	0.761	0.800	0.702	0.699	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	1	0.5	0.745	0.779	0.674	0.698	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	1	0.5	0.734	0.761	0.655	0.709	1.000	1.000	1.000	1.00	1.00	1.00	0.97	0.38
12	1	0.5	0.719	0.737	0.631	0.710	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	1	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.15	0.938	0.973	0.957	0.768	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.15	0.890	0.946	0.911	0.691	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.15	0.852	0.920	0.867	0.657	1.000	1.000	1.000	1.00	1.00	1.00	0.45	0.39
4	1	0.15	0.818	0.894	0.828	0.636	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.40
5	1	0.15	0.791	0.870	0.788	0.630	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.15	0.767	0.845	0.751	0.635	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.15	0.746	0.818	0.708	0.637	1.000	1.000	1.000	1.00	1.00	1.00	0.77	0.40
8	1	0.15	0.730	0.794	0.672	0.646	1.000	1.000	1.000	1.00	0.97	1.00	0.83	0.40
9	1	0.15	0.713	0.767	0.641	0.652	1.000	0.999	1.000	1.00	0.91	1.00	0.88	0.39
10	1	0.15	0.699	0.742	0.609	0.654	1.000	0.995	1.000	1.00	0.86	0.99	0.93	0.39
11	1	0.15	0.686	0.716	0.582	0.664	1.000	0.987	1.000	1.00	0.77	0.92	0.97	0.35
12	1	0.15	0.675	0.692	0.562	0.668	0.988	0.971	0.988	0.75	0.64	0.77	1.00	—
0	1	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	1	0.05	0.936	0.973	0.955	0.759	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0.05	0.887	0.947	0.911	0.686	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0.05	0.844	0.919	0.862	0.645	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.39
4	1	0.05	0.808	0.891	0.818	0.623	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0.05	0.780	0.866	0.775	0.614	1.000	1.000	1.000	1.00	1.00	1.00	0.63	0.40
6	1	0.05	0.755	0.838	0.730	0.620	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0.05	0.732	0.809	0.691	0.620	1.000	0.989	1.000	1.00	0.87	1.00	0.77	0.40
8	1	0.05	0.715	0.782	0.649	0.627	1.000	0.968	1.000	1.00	0.68	1.00	0.83	0.40
9	1	0.05	0.701	0.754	0.615	0.638	1.000	0.920	1.000	1.00	0.45	1.00	0.88	0.39
10	1	0.05	0.689	0.722	0.577	0.643	1.000	0.828	0.999	1.00	0.25	0.92	0.93	0.39
11	1	0.05	0.677	0.685	0.538	0.651	0.998	0.721	0.953	0.92	0.14	0.53	0.97	0.35
12	1	0.05	0.622	0.642	0.506	0.615	0.602	0.593	0.603	0.04	0.03	0.04	1.00	—
0	1	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—

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n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
1	1	0	0.936	0.973	0.955	0.761	1.000	1.000	1.000	1.00	1.00	1.00	0.18	0.39
2	1	0	0.886	0.946	0.910	0.681	1.000	1.000	1.000	1.00	1.00	1.00	0.33	0.39
3	1	0	0.842	0.918	0.860	0.640	1.000	1.000	1.000	1.00	1.00	1.00	0.46	0.38
4	1	0	0.806	0.890	0.819	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.55	0.39
5	1	0	0.779	0.865	0.773	0.613	1.000	1.000	1.000	1.00	1.00	1.00	0.64	0.40
6	1	0	0.754	0.837	0.730	0.619	1.000	1.000	1.000	1.00	1.00	1.00	0.71	0.40
7	1	0	0.731	0.808	0.691	0.620	1.000	0.990	1.000	1.00	0.80	1.00	0.77	0.40
8	1	0	0.713	0.778	0.647	0.627	1.000	0.950	1.000	1.00	0.58	1.00	0.83	0.40
9	1	0	0.698	0.747	0.608	0.637	1.000	0.886	1.000	1.00	0.34	0.99	0.88	0.39
10	1	0	0.689	0.711	0.567	0.645	1.000	0.763	0.999	1.00	0.17	0.93	0.93	0.39
11	1	0	0.677	0.668	0.524	0.650	0.996	0.621	0.928	0.89	0.06	0.44	0.97	0.35
12	1	0	0.606	0.624	0.491	0.601	0.479	0.478	0.478	0.00	0.02	0.00	1.00	—
0	0.75	0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.5	0.958	0.976	0.964	0.855	1.000	1.000	1.000	1.00	1.00	1.00	0.16	0.45
2	0.75	0.5	0.926	0.954	0.931	0.785	1.000	1.000	1.000	1.00	1.00	1.00	0.31	0.44
3	0.75	0.5	0.896	0.930	0.896	0.746	1.000	1.000	1.000	1.00	1.00	1.00	0.43	0.44
4	0.75	0.5	0.866	0.907	0.859	0.722	1.000	1.000	1.000	1.00	1.00	1.00	0.52	0.44
5	0.75	0.5	0.840	0.885	0.825	0.705	1.000	1.000	1.000	1.00	1.00	1.00	0.61	0.45
6	0.75	0.5	0.818	0.863	0.794	0.697	1.000	1.000	1.000	1.00	1.00	1.00	0.69	0.45
7	0.75	0.5	0.796	0.842	0.763	0.690	1.000	1.000	1.000	1.00	1.00	1.00	0.76	0.44
8	0.75	0.5	0.775	0.816	0.735	0.688	1.000	1.000	1.000	1.00	1.00	1.00	0.82	0.44
9	0.75	0.5	0.759	0.794	0.702	0.694	1.000	1.000	1.000	1.00	1.00	1.00	0.87	0.43
10	0.75	0.5	0.742	0.771	0.672	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.92	0.42
11	0.75	0.5	0.723	0.748	0.654	0.696	1.000	1.000	1.000	1.00	1.00	1.00	0.96	0.44
12	0.75	0.5	0.712	0.729	0.633	0.703	1.000	1.000	1.000	1.00	1.00	1.00	1.00	—
0	0.75	0.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.15	0.942	0.972	0.957	0.799	1.000	0.998	1.000	1.00	0.99	1.00	0.17	0.42
2	0.75	0.15	0.898	0.946	0.918	0.718	1.000	0.998	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.15	0.858	0.919	0.874	0.674	1.000	0.997	1.000	1.00	0.98	1.00	0.44	0.41
4	0.75	0.15	0.827	0.892	0.834	0.657	1.000	0.996	1.000	1.00	0.98	1.00	0.54	0.41
5	0.75	0.15	0.795	0.865	0.790	0.641	1.000	0.993	1.000	1.00	0.94	1.00	0.62	0.42
6	0.75	0.15	0.770	0.839	0.756	0.636	1.000	0.986	1.000	1.00	0.89	1.00	0.70	0.42
7	0.75	0.15	0.748	0.815	0.722	0.635	1.000	0.975	1.000	1.00	0.85	1.00	0.77	0.41
8	0.75	0.15	0.729	0.787	0.690	0.639	1.000	0.969	1.000	1.00	0.80	0.99	0.83	0.42
9	0.75	0.15	0.716	0.760	0.652	0.649	0.999	0.952	0.999	0.98	0.72	0.98	0.88	0.41
10	0.75	0.15	0.701	0.737	0.618	0.655	0.997	0.938	0.995	0.94	0.63	0.93	0.93	0.40
11	0.75	0.15	0.688	0.712	0.602	0.659	0.985	0.915	0.984	0.86	0.55	0.80	0.96	0.41
12	0.75	0.15	0.672	0.685	0.571	0.663	0.945	0.893	0.945	0.60	0.44	0.59	1.00	—
0	0.75	0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0.05	0.941	0.972	0.956	0.798	1.000	0.993	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0.05	0.895	0.945	0.913	0.712	1.000	0.992	1.000	1.00	0.99	1.00	0.32	0.41
3	0.75	0.05	0.855	0.916	0.869	0.672	1.000	0.989	1.000	1.00	0.94	1.00	0.44	0.41

Continued on the next page

Continued from the previous page

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
4	0.75	0.05	0.821	0.890	0.830	0.650	1.000	0.981	1.000	1.00	0.89	1.00	0.54	0.41
5	0.75	0.05	0.789	0.861	0.786	0.631	1.000	0.973	1.000	1.00	0.78	1.00	0.63	0.41
6	0.75	0.05	0.767	0.836	0.749	0.633	1.000	0.943	1.000	1.00	0.72	1.00	0.70	0.41
7	0.75	0.05	0.744	0.807	0.707	0.633	1.000	0.919	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0.05	0.723	0.776	0.674	0.634	1.000	0.885	0.999	1.00	0.43	0.96	0.83	0.41
9	0.75	0.05	0.709	0.744	0.631	0.642	0.993	0.802	0.991	0.98	0.28	0.89	0.88	0.41
10	0.75	0.05	0.693	0.713	0.593	0.646	0.968	0.723	0.958	0.92	0.13	0.77	0.93	0.40
11	0.75	0.05	0.674	0.680	0.565	0.644	0.890	0.638	0.877	0.76	0.06	0.61	0.96	0.41
12	0.75	0.05	0.630	0.639	0.524	0.622	0.589	0.527	0.589	0.04	0.01	0.05	1.00	—
0	0.75	0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.00	1.00	1.00	0.00	—
1	0.75	0	0.941	0.972	0.956	0.796	1.000	0.991	1.000	1.00	0.98	1.00	0.17	0.42
2	0.75	0	0.895	0.946	0.917	0.712	1.000	0.994	1.000	1.00	0.98	1.00	0.32	0.41
3	0.75	0	0.855	0.916	0.871	0.669	1.000	0.990	1.000	1.00	0.95	1.00	0.44	0.41
4	0.75	0	0.822	0.890	0.830	0.646	1.000	0.983	1.000	1.00	0.86	1.00	0.54	0.41
5	0.75	0	0.790	0.861	0.788	0.629	1.000	0.966	1.000	1.00	0.79	1.00	0.63	0.41
6	0.75	0	0.767	0.835	0.748	0.632	1.000	0.933	1.000	1.00	0.67	1.00	0.70	0.41
7	0.75	0	0.745	0.805	0.704	0.631	1.000	0.890	1.000	1.00	0.58	0.99	0.77	0.41
8	0.75	0	0.726	0.774	0.672	0.635	1.000	0.855	0.998	0.99	0.35	0.96	0.83	0.41
9	0.75	0	0.709	0.738	0.629	0.642	0.990	0.775	0.987	0.98	0.22	0.90	0.88	0.41
10	0.75	0	0.693	0.705	0.586	0.646	0.962	0.673	0.948	0.91	0.10	0.75	0.93	0.40
11	0.75	0	0.671	0.669	0.557	0.641	0.866	0.569	0.850	0.76	0.02	0.52	0.96	0.41
12	0.75	0	0.616	0.626	0.515	0.608	0.480	0.457	0.480	0.01	0.00	0.01	1.00	—
0	0.5	0.5	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.5	0.963	0.975	0.961	0.894	1.000	0.972	1.000	1.00	0.80	1.00	0.15	0.51
2	0.5	0.5	0.930	0.951	0.924	0.826	1.000	0.972	1.000	1.00	0.80	1.00	0.28	0.50
3	0.5	0.5	0.901	0.928	0.890	0.785	1.000	0.968	1.000	1.00	0.81	1.00	0.39	0.51
4	0.5	0.5	0.876	0.906	0.853	0.758	1.000	0.972	1.000	1.00	0.81	1.00	0.49	0.51
5	0.5	0.5	0.855	0.883	0.823	0.748	1.000	0.968	1.000	1.00	0.81	1.00	0.58	0.51
6	0.5	0.5	0.833	0.861	0.788	0.735	1.000	0.967	1.000	1.00	0.81	1.00	0.66	0.50
7	0.5	0.5	0.816	0.842	0.761	0.735	1.000	0.969	1.000	1.00	0.82	1.00	0.72	0.52
8	0.5	0.5	0.796	0.821	0.733	0.727	1.000	0.971	1.000	1.00	0.81	1.00	0.78	0.53
9	0.5	0.5	0.781	0.802	0.711	0.728	1.000	0.974	1.000	1.00	0.82	1.00	0.85	0.51
10	0.5	0.5	0.763	0.777	0.678	0.725	1.000	0.975	1.000	1.00	0.82	1.00	0.90	0.51
11	0.5	0.5	0.746	0.756	0.652	0.722	1.000	0.973	1.000	1.00	0.83	1.00	0.96	0.48
12	0.5	0.5	0.732	0.734	0.630	0.723	1.000	0.973	1.000	1.00	0.81	1.00	1.00	—
0	0.5	0.15	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.15	0.954	0.974	0.958	0.862	1.000	0.977	1.000	1.00	0.78	1.00	0.15	0.48
2	0.5	0.15	0.915	0.947	0.917	0.787	1.000	0.972	1.000	1.00	0.78	1.00	0.29	0.48
3	0.5	0.15	0.885	0.925	0.877	0.747	1.000	0.964	1.000	1.00	0.74	1.00	0.40	0.49
4	0.5	0.15	0.857	0.900	0.837	0.720	1.000	0.959	1.000	1.00	0.66	1.00	0.50	0.49
5	0.5	0.15	0.833	0.874	0.796	0.708	1.000	0.944	1.000	0.99	0.62	0.99	0.59	0.48
6	0.5	0.15	0.811	0.850	0.758	0.699	1.000	0.934	1.000	0.99	0.58	0.98	0.67	0.48

Continued on the next page

Continued from the previous page

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
7	0.5	0.15	0.795	0.828	0.728	0.701	0.999	0.917	0.999	0.97	0.54	0.96	0.73	0.49
8	0.5	0.15	0.777	0.804	0.690	0.697	0.985	0.902	0.985	0.94	0.49	0.94	0.79	0.51
9	0.5	0.15	0.760	0.781	0.662	0.698	0.980	0.893	0.980	0.90	0.42	0.89	0.86	0.48
10	0.5	0.15	0.741	0.756	0.635	0.695	0.962	0.868	0.963	0.80	0.36	0.80	0.91	0.49
11	0.5	0.15	0.724	0.730	0.608	0.694	0.936	0.850	0.937	0.64	0.31	0.66	0.96	0.46
12	0.5	0.15	0.706	0.708	0.581	0.693	0.890	0.817	0.890	0.37	0.27	0.37	1.00	—
0	0.5	0.05	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0.05	0.954	0.973	0.956	0.862	1.000	0.970	1.000	1.00	0.76	1.00	0.15	0.48
2	0.5	0.05	0.916	0.946	0.913	0.788	1.000	0.951	1.000	1.00	0.70	1.00	0.29	0.48
3	0.5	0.05	0.885	0.923	0.872	0.749	1.000	0.928	1.000	1.00	0.62	0.99	0.40	0.48
4	0.5	0.05	0.857	0.896	0.829	0.722	1.000	0.911	1.000	1.00	0.55	0.99	0.50	0.48
5	0.5	0.05	0.833	0.869	0.788	0.710	0.997	0.885	0.997	0.99	0.48	0.98	0.59	0.48
6	0.5	0.05	0.808	0.841	0.750	0.698	0.997	0.856	0.996	0.98	0.38	0.96	0.67	0.48
7	0.5	0.05	0.791	0.818	0.720	0.697	0.987	0.824	0.986	0.96	0.31	0.94	0.73	0.49
8	0.5	0.05	0.772	0.790	0.678	0.691	0.966	0.785	0.965	0.94	0.25	0.89	0.79	0.51
9	0.5	0.05	0.754	0.764	0.644	0.691	0.946	0.730	0.944	0.89	0.18	0.86	0.86	0.48
10	0.5	0.05	0.731	0.732	0.610	0.684	0.892	0.670	0.891	0.77	0.10	0.74	0.91	0.49
11	0.5	0.05	0.703	0.698	0.577	0.672	0.787	0.621	0.787	0.53	0.06	0.50	0.96	0.46
12	0.5	0.05	0.664	0.668	0.547	0.650	0.561	0.568	0.561	0.02	0.02	0.01	1.00	—
0	0.5	0	1.000	1.000	1.000	1.000	1.000	0.972	1.000	1.00	0.79	1.00	0.00	—
1	0.5	0	0.954	0.973	0.956	0.860	1.000	0.970	1.000	1.00	0.78	1.00	0.16	0.48
2	0.5	0	0.916	0.946	0.913	0.787	1.000	0.948	1.000	1.00	0.68	1.00	0.29	0.47
3	0.5	0	0.884	0.921	0.870	0.747	1.000	0.908	1.000	1.00	0.64	1.00	0.40	0.48
4	0.5	0	0.855	0.894	0.826	0.717	1.000	0.885	1.000	1.00	0.57	1.00	0.50	0.48
5	0.5	0	0.831	0.866	0.785	0.706	0.996	0.853	0.996	0.99	0.46	0.99	0.59	0.48
6	0.5	0	0.806	0.838	0.745	0.694	0.996	0.809	0.994	0.98	0.37	0.97	0.67	0.48
7	0.5	0	0.789	0.815	0.715	0.694	0.985	0.773	0.983	0.96	0.30	0.94	0.73	0.49
8	0.5	0	0.771	0.785	0.673	0.689	0.966	0.737	0.964	0.94	0.22	0.90	0.79	0.51
9	0.5	0	0.753	0.757	0.639	0.689	0.946	0.670	0.943	0.89	0.12	0.84	0.86	0.48
10	0.5	0	0.729	0.723	0.604	0.683	0.890	0.611	0.888	0.77	0.08	0.70	0.91	0.49
11	0.5	0	0.699	0.688	0.569	0.667	0.760	0.537	0.757	0.52	0.04	0.47	0.96	0.46
12	0.5	0	0.653	0.655	0.536	0.639	0.488	0.483	0.488	0.02	0.03	0.02	1.00	—
0	0.25	0.5	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.5	0.978	0.981	0.967	0.949	1.000	0.769	1.000	0.98	0.30	0.99	0.12	0.62
2	0.25	0.5	0.956	0.962	0.938	0.910	1.000	0.764	1.000	0.98	0.29	0.99	0.24	0.61
3	0.25	0.5	0.939	0.946	0.907	0.882	1.000	0.765	1.000	0.98	0.30	0.98	0.34	0.62
4	0.25	0.5	0.922	0.928	0.870	0.864	1.000	0.761	1.000	0.97	0.32	0.97	0.44	0.61
5	0.25	0.5	0.905	0.910	0.844	0.846	1.000	0.762	1.000	0.97	0.32	0.97	0.53	0.60
6	0.25	0.5	0.889	0.892	0.818	0.834	1.000	0.772	1.000	0.96	0.32	0.96	0.61	0.61
7	0.25	0.5	0.876	0.877	0.792	0.825	0.998	0.780	0.998	0.94	0.30	0.95	0.69	0.60
8	0.25	0.5	0.861	0.859	0.766	0.817	0.998	0.779	0.998	0.93	0.29	0.93	0.76	0.62
9	0.25	0.5	0.842	0.837	0.740	0.806	0.998	0.771	0.998	0.92	0.28	0.92	0.82	0.62

Continued on the next page

Continued from the previous page

n_{bad}	α	SNR	ρ mean	ρ rank	ρ med	ρ IVW	AUC mean	AUC med	AUC IVW	P mean	P med	P IVW	w_{bad}	σ^2 ratio
10	0.25	0.5	0.827	0.819	0.715	0.799	0.998	0.772	0.999	0.92	0.29	0.92	0.89	0.60
11	0.25	0.5	0.815	0.802	0.691	0.800	0.998	0.772	0.998	0.91	0.30	0.91	0.94	0.67
12	0.25	0.5	0.802	0.783	0.671	0.797	0.998	0.769	0.998	0.91	0.30	0.90	1.00	—
0	0.25	0.15	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.15	0.973	0.979	0.964	0.938	0.991	0.765	0.991	0.98	0.29	0.98	0.13	0.61
2	0.25	0.15	0.949	0.960	0.933	0.893	0.990	0.758	0.989	0.98	0.23	0.98	0.25	0.60
3	0.25	0.15	0.929	0.942	0.900	0.862	0.984	0.745	0.983	0.97	0.21	0.97	0.35	0.61
4	0.25	0.15	0.909	0.922	0.864	0.844	0.979	0.741	0.979	0.95	0.21	0.95	0.45	0.60
5	0.25	0.15	0.888	0.902	0.835	0.823	0.968	0.734	0.969	0.92	0.19	0.92	0.54	0.60
6	0.25	0.15	0.874	0.883	0.804	0.812	0.953	0.738	0.953	0.86	0.18	0.86	0.61	0.61
7	0.25	0.15	0.856	0.866	0.779	0.801	0.930	0.733	0.929	0.79	0.13	0.79	0.69	0.60
8	0.25	0.15	0.837	0.847	0.750	0.789	0.905	0.710	0.907	0.72	0.14	0.72	0.76	0.61
9	0.25	0.15	0.819	0.824	0.717	0.780	0.880	0.697	0.882	0.65	0.10	0.67	0.83	0.61
10	0.25	0.15	0.801	0.802	0.690	0.770	0.857	0.674	0.859	0.52	0.10	0.54	0.89	0.59
11	0.25	0.15	0.785	0.784	0.662	0.767	0.799	0.657	0.800	0.37	0.08	0.37	0.94	0.66
12	0.25	0.15	0.768	0.765	0.639	0.760	0.744	0.646	0.744	0.19	0.05	0.18	1.00	—
0	0.25	0.05	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0.05	0.973	0.978	0.963	0.936	0.988	0.761	0.986	0.98	0.26	0.98	0.13	0.61
2	0.25	0.05	0.949	0.958	0.929	0.891	0.987	0.727	0.984	0.98	0.17	0.98	0.25	0.59
3	0.25	0.05	0.928	0.941	0.898	0.859	0.980	0.710	0.978	0.97	0.15	0.97	0.35	0.60
4	0.25	0.05	0.908	0.920	0.861	0.840	0.974	0.690	0.972	0.95	0.12	0.95	0.45	0.60
5	0.25	0.05	0.886	0.897	0.830	0.819	0.956	0.679	0.953	0.92	0.12	0.92	0.54	0.59
6	0.25	0.05	0.869	0.876	0.798	0.806	0.927	0.660	0.923	0.84	0.09	0.84	0.62	0.60
7	0.25	0.05	0.852	0.859	0.771	0.797	0.895	0.639	0.893	0.76	0.07	0.77	0.69	0.60
8	0.25	0.05	0.832	0.838	0.740	0.783	0.858	0.610	0.856	0.67	0.06	0.68	0.76	0.61
9	0.25	0.05	0.812	0.813	0.705	0.773	0.811	0.575	0.812	0.58	0.04	0.58	0.83	0.61
10	0.25	0.05	0.791	0.791	0.676	0.760	0.761	0.534	0.763	0.47	0.02	0.47	0.89	0.59
11	0.25	0.05	0.766	0.767	0.645	0.748	0.637	0.512	0.639	0.27	0.02	0.27	0.94	0.66
12	0.25	0.05	0.741	0.744	0.620	0.734	0.519	0.487	0.522	0.01	0.02	0.01	1.00	—
0	0.25	0	1.000	1.000	1.000	1.000	1.000	0.768	1.000	1.00	0.29	1.00	0.00	—
1	0.25	0	0.973	0.978	0.964	0.935	0.987	0.757	0.985	0.98	0.27	0.98	0.13	0.61
2	0.25	0	0.948	0.958	0.929	0.890	0.986	0.726	0.983	0.98	0.18	0.98	0.25	0.59
3	0.25	0	0.928	0.940	0.897	0.858	0.979	0.697	0.977	0.97	0.13	0.97	0.35	0.60
4	0.25	0	0.907	0.918	0.857	0.839	0.973	0.670	0.970	0.95	0.11	0.95	0.45	0.59
5	0.25	0	0.886	0.897	0.828	0.819	0.954	0.648	0.950	0.92	0.08	0.92	0.54	0.59
6	0.25	0	0.869	0.874	0.795	0.807	0.922	0.627	0.918	0.83	0.07	0.83	0.62	0.60
7	0.25	0	0.852	0.858	0.770	0.797	0.888	0.602	0.885	0.76	0.06	0.76	0.69	0.60
8	0.25	0	0.830	0.835	0.738	0.781	0.844	0.564	0.843	0.67	0.02	0.67	0.76	0.61
9	0.25	0	0.810	0.810	0.701	0.770	0.791	0.527	0.792	0.58	0.03	0.58	0.83	0.61
10	0.25	0	0.789	0.788	0.673	0.758	0.741	0.503	0.744	0.46	0.02	0.45	0.89	0.59
11	0.25	0	0.763	0.764	0.642	0.744	0.610	0.476	0.611	0.25	0.02	0.25	0.94	0.66
12	0.25	0	0.736	0.741	0.617	0.728	0.480	0.455	0.482	0.00	0.00	0.00	1.00	—

Table 12: Sample-size imbalance (Dirichlet α) and mixed sites. Spearman ρ of the observed aggregate against the clean aggregate. $B = 10$, $p = 80$, $K = 10$, $N = 4000$, five replicates.

α	Regime	Uniform	n-weighted	IVW	Median
0.05	Covariate	1.000	1.000	0.788	1.000
0.05	Concept	0.996	0.913	0.876	0.975
0.1	Covariate	1.000	1.000	0.754	1.000
0.1	Concept	0.996	0.921	0.888	0.985
1	Covariate	1.000	1.000	0.883	1.000
1	Concept	0.991	0.952	0.738	0.964
10	Covariate	1.000	1.000	0.971	1.000
10	Concept	0.984	0.965	0.610	0.961
20	Covariate	1.000	1.000	0.977	1.000
20	Concept	0.985	0.976	0.749	0.950
Mixed 5 CS + 5 CD		0.300	0.300	—	0.295
Mixed (vs. all-CD)		0.334	0.334	—	0.372

Sample-size imbalance is a different corruption. Table 12 reports Spearman correlation of the aggregate with the clean aggregate under a Dirichlet allocation of site sizes. Uniform weights and the median stay at 1 for covariate shift across the concentration grid, including the severe imbalance at concentration 0.05. Inverse-variance weights are lower on that grid, down to 0.610 for concept drift at concentration 10. A pool of five covariate-shift sites and five concept-drift sites correlates only about 0.30 with the clean aggregate, for every aggregator that is defined. Size weights do not turn two kinds of shift into one ranking. That is the gap filled by the site score in Section 1.5.

On injected shifts in SECOM [13], Diabetes130 [14], TEP [4], and TCGA [20], mean share keeps a high selection AUC at five corrupted centers out of the reported grid (Table 13). At ten corrupted centers the ranking moves: SECOM mean-share AUC falls to 0.708 and TEP to 0.580, while TCGA stays at 0.977. The median is not uniformly more fragile on this table. At ten corrupted SECOM centers its Spearman is 0.758 against 0.657 for the mean. The high-dimensional synthetic grid and these four data sets therefore agree on one fact and not on another. A moderate number of corrupted centers leaves the selected set intact when a clean signal remains. Which of the mean and the median wins the rank correlation depends on the design. On the same two data sets, a pooled comparison of the selectors themselves is Table 14. The FSDS row is marked in bold. Degree and the random forest dominate covariate AUC. Under concept drift the ordering changes: on Diabetes130 with a random split, the knockoff reaches concept AUC 0.996, while FSDS is 0.742.

A homogeneous synthetic check uses $d = 10$, $n = 2400$, and $K = 6$ blocks, three shifted and three clean, with equal sizes or a Dirichlet(0.5) allocation. The regression on the clean blocks is linear in the first five coordinates plus standard normal noise. Covariate shifts are a mean shift, a covariance shift, or a mixture shift. Concept drifts added to the regression are linear, sine, cubic, interaction, and threshold. Across that list the Spearman correlation between the merged vector and the pooled vector stays in $[0.79, 0.93]$, averaged over two seeds and both size allocations. The highest entry is linear drift with sample-size weights (0.933). The lowest is the uniform average under the product interaction (0.803). With no corruption, a mixed block of two covariate-shift clients, two concept-drift clients, one mixed client, and one clean client gives Spearman 0.838 (uniform), 0.867 (n-weighted), 0.842 (median), and 0.879 (client-agreement Shapley) against the centralized importance. Taking the top three or top five clients and then applying sample-size weights raises that to 0.925 and 0.929. Screening clients before averaging is the same idea as the Δ^{cf} weight, computed here by agreement with the central vector rather than by a local skill score. Table 15 is the homogeneous list. Table 16 is the mixed block with no attack.

1.3 Real data: images, WhyShift, and TabReD

FEMNIST [3] and WhyShift [11] check ranking consistency when the reference is the pooled vector on the same images or the same table. For a character pair, a 30-component PCA is fit on the pooled pair and each batch is cut into $K \in \{5, 10, 20, 30\}$ blocks. The three uploaded vectors are random-forest domain importance, leave-one-coordinate RBF MMD, and the increase in pseudo-outcome risk under conditional permutation. Table 17 is the sample-size weighted merge against the pooled wrapper on character pairs. Table 18 is the same comparison after the pair is treated as a covariate shift, averaged over the four adjacent pairs among the five most frequent characters. With a common cap of 800 images per class, the sample-size weighted mean

Table 13: Federated alignment on SECOM, TCGA, Diabetes130, and TEP under concept drift. AUC, precision at K, and Spearman ρ at $n_{\text{bad}} \in \{0, 5, 10\}$.

		Mean share			Median share		
Dataset	n_{bad}	AUC	$P@K$	ρ	AUC	$P@K$	ρ
SECOM	0	1.000	1.00	1.000	1.000	1.00	1.000
SECOM	5	1.000	0.98	0.894	0.995	0.92	0.917
SECOM	10	0.708	0.40	0.657	0.683	0.34	0.758
TCGA	0	1.000	1.00	1.000	1.000	1.00	1.000
TCGA	5	1.000	1.00	0.869	0.999	0.98	0.877
TCGA	10	0.977	0.84	0.771	0.974	0.82	0.758
Diabetes	0	1.000	1.00	1.000	1.000	1.00	1.000
Diabetes	5	1.000	1.00	0.887	1.000	1.00	0.796
Diabetes	10	0.920	0.75	0.814	0.940	0.85	0.635
TEP	0	1.000	1.00	1.000	1.000	1.00	1.000
TEP	5	0.996	0.88	0.868	0.970	0.68	0.728
TEP	10	0.580	0.12	0.582	0.586	0.20	0.581

Table 14: Pooled comparison on SECOM and Diabetes130 under covariate (Cov) and concept (Con) shift. AUC and precision at 10. Random split and `max_feature` split. The FSDS row is in bold. PermuCATE on SECOM random covariate is the mean of two replicates; one replicate had a one-class label batch and the fit was skipped.

	SECOM (random)				Diabetes130 (random)				SECOM (`max_feature`)				Diabetes130 (`max_feature`)			
Method	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$	Cov AUC	Cov $P@10$	Con AUC	Con $P@10$
PO	0.452	0.400	0.619	0.600	0.308	0.633	0.892	0.867	0.487	0.267	0.579	0.600	0.492	0.733	0.758	0.767
FSDS	**0.716**	**0.433**	**0.598**	**0.400**	**0.792**	**0.867**	**0.742**	**0.800**	**0.650**	**0.267**	**0.442**	**0.200**	**0.483**	**0.733**	**0.713**	**0.767**
deg	0.950	0.833	0.552	0.333	0.992	0.967	0.467	0.733	0.916	0.767	0.429	0.167	0.850	0.900	0.583	0.733
RF	1.000	1.000	0.549	0.300	1.000	1.000	0.558	0.800	0.920	0.733	0.459	0.200	0.917	0.967	0.617	0.767
RF-LOCO	0.622	0.367	0.447	0.267	0.758	0.767	0.475	0.733	0.631	0.367	0.418	0.200	0.575	0.767	0.433	0.667
LOCO-MMD	0.987	0.967	0.496	0.233	0.983	0.967	0.475	0.733	0.897	0.800	0.517	0.233	0.700	0.767	0.408	0.700
HSIC-LOCO	0.402	0.167	0.577	0.367	0.575	0.767	0.475	0.767	0.505	0.367	0.612	0.367	0.558	0.767	0.475	0.700
GRF	0.543	0.267	0.774	0.633	0.442	0.700	0.692	0.833	0.526	0.300	0.667	0.433	0.433	0.700	0.592	0.733
ℓ_1 mis	0.580	0.267	0.803	0.667	0.458	0.700	0.725	0.767	0.718	0.500	0.697	0.500	0.533	0.767	0.700	0.767
ℓ_1 spec	0.587	0.367	0.796	0.633	0.583	0.767	0.550	0.700	0.657	0.467	0.749	0.567	0.550	0.733	0.683	0.767
ΔSHAP	0.587	0.267	0.778	0.633	0.475	0.733	0.750	0.800	0.478	0.233	0.744	0.600	0.333	0.633	0.792	0.800
knockoff	0.578	0.367	0.796	0.600	0.500	0.733	0.996	0.967	0.626	0.367	0.749	0.633	0.329	0.667	0.883	0.900
PermuCATE	0.523	0.350	0.748	0.600	0.492	0.733	0.808	0.800	0.539	0.367	0.709	0.567	0.450	0.667	0.717	0.767

coincides with the uniform mean. Pseudo-outcome risk is omitted from that table: the character label is the batch indicator, so the pseudo-outcome does not depend on X. As a direct prediction problem, a random forest classifies digits 0–9 from 30 pooled PCA scores on 200,000 images. Importance is the drop in accuracy when a component is permuted. Twenty blocks of 10,000 have equal size, so the uniform average and the sample-size average coincide. Their Spearman correlation with the pooled importance is 0.996 (Pearson 0.996). Inverse-variance weighting yields 0.993. A single block, before merging, already correlates at 0.975, and the first four components are $2, 3, 6, 9$ in both the pooled vector and the merge. Information loss from cutting the sample is small when every block is a copy of the same prediction problem.

Pseudo-outcome risk on the same images does not follow this pattern. With a balanced random treatment arm and the digit as the outcome, the pooled PO-risk vector has mean absolute entry 0.0007. Inverse-variance and sample-size merges correlate with it at -0.02 and -0.01. The treatment is independent of the image, so the wrapper has no residual contrast. The WhyShift folds in Table 3 are the tabular version of the same split: covariate wrappers aggregate, and PO-risk does not.

TabReD is the balanced tabular check. The largest site fraction on the summary grid is 0.24, so Table 1 predicts a small gain for $n|S|$. The mean Spearman values are 0.453 (uniform), 0.460 (sample size), 0.466 ($n|S|$), and 0.453 (rank). $n|S|$ is a safe default because it ties the uniform average.

A second TabReD partition scores the aggregate against whole-batch FSDS. The splits are not those of Table 19. Cooking `num_184` repeats the `num_105` geometry: pairwise concordance 0.875, and every convex weight correlates only about 0.12 with the whole batch. Weather `num_5` and `num_84` have pairwise concordance near 0.78 and a negative Spearman for every convex weight. Weather `num_98` has the lowest pairwise concordance on that table (0.440) and the highest $n|S|$ correlation (0.631).

Single-batch F1 and AUC against the top linear coefficients are a method comparison, not a federated metric. Mean F1 is 0.433 for Delta-Shapley, 0.400 for PermuCATE and for LOCO, 0.367 for the generalized

Table 15: Homogeneous shifts on six clients, three shifted and three clean. Spearman correlation of the merged permutation importance with the centralized importance. Entries are averages over two seeds and both size allocations (equal sizes, or Dirichlet0.5). (δ_X, δ_C) is the covariate shift and the concept drift.

Shift	(δ_X, δ_C)	Uniform	n-weighted	IVW	Median
None	$(0, 0)$	0.891	0.900	0.903	0.885
Mean	$(0.5, 0)$	0.815	0.794	0.791	0.815
Mean	$(1.5, 0)$	0.839	0.827	0.833	0.861
Covariance	$(0.6, 0)$	0.842	0.836	0.833	0.845
Mixture	$(0.8, 0)$	0.809	0.845	0.848	0.848
Linear CD	$(0, 1)$	0.903	0.933	0.921	0.897
Nonlinear CD	$(0, 1)$	0.867	0.894	0.888	0.873
Cubic CD	$(0, 0.4)$	0.842	0.845	0.852	0.888
Interaction CD	$(0, 1)$	0.803	0.830	0.855	0.830
Threshold CD	$(0, 1)$	0.867	0.867	0.873	0.873
Mean + nonlinear	$(1, 1)$	0.836	0.830	0.821	0.848
Mean + interaction	$(1, 1)$	0.836	0.848	0.845	0.909

Table 16: Mixed blocks with no attack: two covariate-shift clients, two concept-drift clients, one mixed client, and one clean client. Correlation with the centralized importance.

Aggregator	Spearman	Pearson
Uniform	0.838	0.920
n-weighted	0.867	0.952
Median	0.842	0.921
Shapley (client agreement)	0.879	0.922
Top-3, then n-weighted	0.925	0.957
Top-5, then n-weighted	0.929	0.951

random forest, 0.333 for the forest, 0.311 for FSDS, and 0.178 for the knockoff. Mean selection AUC has the same lead: Delta-Shapley 0.701, the generalized random forest 0.677, the forest 0.647, FSDS 0.588. FSDS and the forest take 0.6 seconds per cell. PermuCATE, LOCO, and the generalized random forest share one causal block of 56 seconds; the column total 220.5 adds that block three times.

Leave-one-site-out influence holds the whole-batch vector fixed and deletes one upload:

$$\Delta_k = \rho(\text{agg}, \text{whole}) - \rho(\text{agg}_{-k}, \text{whole}).$$

On the random partition, $\rho_{n|S|} = 0.889 \pm 0.106$ over 45 cells and the largest $|\Delta_k|$ averages 0.054. On the feature-rank partition, $\rho_{n|S|} = 0.243 \pm 0.241$ and the largest $|\Delta_k|$ averages 0.057. The deletion moves the aggregate by the same amount whether the aggregate is aligned or not. Weather `num_98`, the strongest feature-rank cell ($\rho = 0.645$), moves by 0.093. Weather `num_5`, a negative cell, moves by 0.031. Influence does not rank a misaligned split above an aligned one, because deleting one site cannot reveal that every site points the same wrong way.

1.4 Statistical justification

Write $V_{j,k}$ for the importance of feature j at site k. Dependence across features inside a site is unrestricted. Across sites the uploads are treated as independent. This is a statement about the uploaded vectors, not about multi-modal attribution with several loaders. The guarantee uses consistency of the local estimator in mean and variance: $\hat{V}_{j,k} \xrightarrow{p} V^{\star}_{j,k}$ and $\hat{\sigma}^2_{j,k} \xrightarrow{p} \sigma^2_{j,k}$. Normalizing each upload onto the simplex gives $\|V_k\|_2 \le \|V_k\|_1 = 1$, so the aggregate converges to its expectation, and the Matrix Bernstein inequality upgrades pointwise consistency to a uniform bound over the p coordinates. The bound does not require the $p \times p$ covariance of the coordinates. The dimension enters as $\log(2p/\delta)$.

Theorem (Matrix Bernstein bound for aggregated VIMP). Let $V_k \in [0, 1]^p$ with $\sum_{j=1}^{p} V_{k,j} = 1$ for each $k \in \{1, \dots, K\}$, and let $\bar{V} = K^{-1} \sum_{k=1}^{K} V_k$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \bar{V} - \mathbb{E}[\bar{V}] \right\|_2 \le \sqrt{\frac{2 \log(2p/\delta)}{K}} + \frac{2 \log(2p/\delta)}{3K}. \tag{2}$$

Table 17: FEMNIST character pairs. Spearman correlation of the n-weighted merge with the pooled out-of-distribution wrapper.

Wrapper output	$K = 5$	$K = 10$	$K = 20$	$K = 30$
RF domain classifier	0.795	0.819	0.843	0.830
MMD-LOCO	0.781	0.887	0.903	0.909

Table 18: FEMNIST covariate-shift attribution. Spearman correlation with the pooled ranking, averaged over the four adjacent pairs among the five most frequent characters (1–7, 7–3, 3–0, 0–2). Each pair is projected to 30 principal components fit on the pooled images. Every site reuses those coordinates and uses at most 800 images per class. Single-site is the mean correlation of the site-level rankings. Mean and median are the coordinate-wise aggregates. The sample-size weighted mean coincides with the unweighted mean under that cap.

Sites	Method	Single-site	Mean	Median
5	RF domain	0.74	0.80	0.78
5	MMD-LOCO	0.79	0.78	0.81
10	RF domain	0.73	0.82	0.81
10	MMD-LOCO	0.83	0.89	0.89
20	RF domain	0.74	0.84	0.84
20	MMD-LOCO	0.84	0.90	0.90
30	RF domain	0.74	0.83	0.83
30	MMD-LOCO	0.83	0.91	0.88

The same simplex bound is the starting point for a mean, a median, and inverse-variance weights. The median and the mean satisfy it once a consistent variance estimator exists, for example from the bootstrap, the jackknife, or the bag of little bootstraps. Inverse-variance weights need the additional lower bound in Assumption (A6). If that bound fails for some site, truncated weights or equal weights restore the coordinate-wise inequality below. Each site uploads its normalized vector once. Relative to iterative synchronization of a high-dimensional model, the communication is smaller by a factor of order $1/K$, and no raw row is sent.

Claim. Under Assumptions (A1)–(A4), the one-shot aggregate obeys the scalar and coordinate-wise bounds (3)–(4). Under the gap condition (A5), the empirical Top-K set of $\bar{V}$ equals that of $\mathbb{E}\bar{V}$ on the event $\{\|\bar{V} - \mathbb{E}\bar{V}\|_\infty < \text{margin}/2\}$. Under (A6), the leading range factor in (4) may be replaced by an empirical variance proxy. Any simplex upload satisfies the assumptions. They do not require a low-rank or sparse model for $\Sigma = \mathrm{Cov}(V_k)$. The inequalities are not asserted to dominate the spectral bound for every Σ, and they are not asserted to be minimax. They are enough for coordinate-wise control and for a Top-K corollary.

Assumptions.

(A1) $V_k \in [0,1]^p$ with $\mathbf{1}^\top V_k = 1$; the sites are independent; $\bar{V} = \sum_k w_k V_k$ with $w_k \geq 0$ and $\sum_k w_k = 1$.

(A2) $\hat{R} := \max_k \|V_k\|_2$ and $R_\star \geq \hat{R}$ (for example $R_\star = 1$).

(A3) K_{eff} is the effective sample size induced by (w_k). It equals K when $w_k = 1/K$.

(A4) A fixed contrast $a \in \mathbb{R}^p$ with $\|a\|_2 \leq 1$; $v_a = \mathrm{Var}(a^\top V_1)$ under identical site laws, or the analogous weighted variance otherwise.

(A5) The population shares separate a designated index set of size K from its complement by a margin margin > 0.

(A6) Consistent estimates $\hat{\sigma}_k^2 \geq \underline{\sigma}^2 > 0$ for the coordinates or the contrast in use.

Dependence among coordinates inside a site is unrestricted, and Σ is not estimated. No condition is imposed on $\mathrm{rank}(\Sigma)$. In particular, (A1) and (A2) hold for every FSDS share.

Table 19: TabReD federated FSDS on nine task×split cells. Site sizes are nearly balanced (max-frac = 0.24). Cooking `num_105` agrees with itself and disagrees with the whole batch.

task	split	pair	site↔whole	unif	n	$n\|S\|$	rank	best
cooking	num_105	0.884	−0.198	−0.200	−0.191	−0.204	−0.209	n
cooking	num_80	0.383	0.161	0.250	0.305	0.329	0.251	$n\|S\|$
cooking	num_69	0.835	0.468	0.520	0.525	0.530	0.520	$n\|S\|$
ecom	num_109	0.787	0.545	0.642	0.645	0.647	0.635	$n\|S\|$
ecom	num_100	0.825	0.532	0.598	0.596	0.610	0.605	$n\|S\|$
ecom	num_103	0.851	0.470	0.498	0.505	0.516	0.501	$n\|S\|$
weather	num_38	0.598	0.428	0.554	0.514	0.531	0.546	unif
weather	num_44	0.716	0.725	0.836	0.840	0.846	0.827	$n\|S\|$
weather	num_37	0.519	0.283	0.378	0.399	0.389	0.401	med
mean	—	0.711	0.379	0.453	0.460	0.466	0.453	—

Table 20: Cooking-time `num_105`: feature-rank versus random array-split (mean±SD of ρ). Aggregator headroom is about zero on both protocols.

protocol	n	pair	unif	n	$n\|S\|$	Δ
feature-rank	20	0.88 ± 0.01	0.07 ± 0.17	0.07 ± 0.17	0.07 ± 0.17	+0.01
random	200	0.26 ± 0.06	0.80 ± 0.06	0.81 ± 0.06	0.80 ± 0.06	+0.00

Spectral baseline. Under (A1)–(A3) and a bound R on $\|w_k(V_k - \mathbb{E}V_k)\|_2$, Matrix Bernstein [15, 12] yields, with probability at least $1 - \delta$,

$$\|\bar{V} - \mathbb{E}\bar{V}\|_2 \le C\Big(R\sqrt{\frac{\log(2p/\delta)}{K_{\text{eff}}}} + R\frac{\log(2p/\delta)}{K_{\text{eff}}}\Big).$$

The substitution used below is $R \leftarrow R_\star$ from (A2).

Scalar and ℓ_∞ bounds. Under (A1)–(A4), with probability at least $1 - \delta$,

$$\big|a^\top(\bar{V} - \mathbb{E}\bar{V})\big| \le C\Big(\sqrt{\frac{v_a \log(2/\delta)}{K_{\text{eff}}}} + R_\star\frac{\log(2/\delta)}{K_{\text{eff}}}\Big), \tag{3}$$

and, by a union bound over coordinates,

$$\|\bar{V} - \mathbb{E}\bar{V}\|_\infty \le C\Big(R_\star\sqrt{\frac{\log(2p/\delta)}{K_{\text{eff}}}} + R_\star\frac{\log(2p/\delta)}{K_{\text{eff}}}\Big). \tag{4}$$

Equation (4) is the deviation that enters the Top-K event. It is not presented as a strict improvement of the spectral-norm inequality for every covariance.

Top-K corollary. Under (A1)–(A5), the event $\{\|\bar{V} - \mathbb{E}\bar{V}\|_\infty < \text{margin}/2\}$ implies equality of the empirical and population Top-K sets. Together with (4), a sufficient condition is

$$C\Big(R_\star\sqrt{\frac{\log(2p/\delta)}{K_{\text{eff}}}} + R_\star\frac{\log(2p/\delta)}{K_{\text{eff}}}\Big) < \frac{1}{2}\,\text{margin}.$$

If the margin lies below this threshold, Top-K instability comes from insufficient separation of the population shares, not from the choice among the mean, the median, and inverse-variance weights. Concordance of the full Spearman ranking is not claimed.

Table 21: TabReD federated FSDS. Primary metric: Spearman correlation of the aggregate with whole-batch FSDS.

task	split	pair	ρ_{unif}	ρ_n	$\rho_{n\|S\|}$	ρ_{rank}	best
cooking-time	num_184	0.875	0.114	0.111	0.127	0.119	$n\|S\|$
cooking-time	num_38	0.885	0.399	0.403	0.400	0.393	n
cooking-time	num_123	0.812	0.403	0.433	0.433	0.397	$n\|S\|$
ecom-offers	num_67	0.769	0.321	0.362	0.356	0.300	n
ecom-offers	num_104	0.768	0.222	0.222	0.233	0.215	$n\|S\|$
ecom-offers	num_60	0.649	0.099	0.106	0.107	0.071	$n\|S\|$
weather	num_5	0.791	−0.054	−0.061	−0.057	−0.064	median
weather	num_84	0.767	−0.111	−0.118	−0.108	−0.131	median
weather	num_98	0.440	0.615	0.623	0.631	0.559	$n\|S\|$

Table 22: TabReD, single batch: F1 against the top $|\beta|$ of a linear regression. PermuCATE uses 50 permutations and a random-forest outcome. Not a federated metric.

task	split	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	L1_m	L1_s	KO
cooking-time	num_184	0.100	0.200	0.500	0.200	0.400	0.200	0.200	0.400	0.300	0.200
cooking-time	num_38	0.100	0.200	0.600	0.200	0.400	0.200	0.400	0.400	0.000	0.000
cooking-time	num_123	0.100	0.200	0.300	0.200	0.300	0.200	0.200	0.600	0.000	0.000
ecom-offers	num_67	0.500	0.500	0.400	0.500	0.500	0.500	0.500	0.200	0.200	0.000
ecom-offers	num_104	0.500	0.500	0.400	0.400	0.400	0.500	0.500	0.300	0.300	0.200
ecom-offers	num_60	0.400	0.500	0.200	0.400	0.400	0.500	0.400	0.300	0.400	0.100
weather	num_5	0.400	0.300	0.400	0.200	0.400	0.500	0.200	0.300	0.200	0.400
weather	num_84	0.300	0.300	0.500	0.200	0.300	0.500	0.400	0.300	0.500	0.300
weather	num_98	0.400	0.300	0.600	0.300	0.500	0.500	0.500	0.300	0.300	0.400
mean	—	0.311	0.333	0.433	0.289	0.400	0.400	0.367	0.344	0.244	0.178

Empirical Bernstein. Under (A1)–(A4) and (A6), empirical Bernstein [18, 16] replaces the leading factor $R_\star$ in (4) by a consistent variance proxy, up to a remainder of order $R_\star \log(2p/\delta)/K_{\text{eff}}$. The same variance input is required for inverse-variance weights [5]. If the lower bound in (A6) fails at some site, truncated weights or equal weights restore (4).

The bound concentrates $\bar{V}$ around $\mathbb{E}\bar{V}$. It does not identify which sites belong in that expectation. A noise site and a drifted site can both upload a large vector. The expectation of the uniform average then mixes a concept with a consensus breaker, and the inequality still holds around that mixture. Section 1.5 is the justification for which sites enter the sum.

1.5 Justifying shift drivers across sites

A driver is a site whose conditional distribution $P(Y \mid X)$ has changed, in a way a local model can learn, relative to a shared burn-in. The ten-site design makes that set known. Covariates are the ACS Income and ACS Public Coverage extracts used by WhyShift, California 2018, winsorized and standardized on the analysis sample, then cut at random into $K = 10$ sites of 1600 rows. Every site therefore shares the same empirical $P(X)$. Each site is $T = 16$ batches of 100 rows. The first four batches are the burn-in, still labeled by the real outcome y^{base}. At batch $t^\star = 8$, four clean sites keep y^{base}, three linear sites rewrite an upper tail of a designated coordinate X_1 by a site-specific halfspace whose positive rate matches the baseline, two nonlinear sites rewrite every activated row by a product rule or by $\sin(X_1) + X_3^2$, again rate-matched, and one noise site draws independent Bernoulli(1/2) labels. The noise site destroys consensus. It is not a driver. There is no new concept to learn.

The site score uses a random forest with 25 trees, depth 8, and minimum leaf size 5. Local probabilities on the most recent six batches are three-fold cross-fit. Reference probabilities are the average of two forests, each trained on a subset of the pooled burn-in whose size equals the local training set. On a region $\mathcal{R}$ with both classes present, at least fifty current rows, and at least eighty reference rows,

$$\Delta_k(\mathcal{R}) = \big(B(\mu_0^{\text{sm}}) - B(\mu_k^{\text{oof}})\big)_+ \cdot \big(2(\text{AUC}(\mu_k^{\text{oof}}) - 1/2)\big)_+, \tag{5}$$

Table 23: TabReD: mean selection AUC by task, same run as Table 22.

task	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO
cooking-time	0.537	0.588	0.673	0.500	0.510	0.583	0.596	0.758	0.416	0.319
ecom-offers	0.797	0.792	0.700	0.626	0.642	0.613	0.806	0.511	0.572	0.361
weather	0.430	0.561	0.729	0.588	0.682	0.687	0.629	0.512	0.510	0.536
mean	0.588	0.647	0.701	0.571	0.611	0.628	0.677	0.594	0.499	0.405

Table 24: TabReD wall seconds per cell, nine cells. PermuCATE, LOCO, and GRF share one causal block. The total adds that block three times.

	FSDS	RF	ΔSH	MMD	PC	LOCO	GRF	$L1_m$	$L1_s$	KO	total
sec	0.6	0.6	7.9	20.9	56.0	56.0	56.0	7.5	7.5	7.5	220.5
× vs. first	1.0	1.0	14.2	37.6	100.9	100.9	100.9	13.5	13.5	13.5	—

and the reported score is

$$\Delta_k^{\mathrm{cf}} = \max_{\mathcal{R} \in \{\mathrm{all},\, \mathrm{high},\, \mathrm{low}\}} \Delta_k(\mathcal{R}). \tag{6}$$

High and low are the two sides of the burn-in median of X_1. The first factor is the excess Brier risk of the size-matched reference. The second factor is zero when the local model does not beat a coin. The concept contrast on the winning region is $\tau_k^2 = \mathrm{mean}(\mu_k^{\mathrm{oof}} - \mu_0^{\mathrm{sm}})^2$, and $\tau_k^{2,\mathrm{g}}$ multiplies it by the same skill factor. Pairwise concordance $C(s; \mathcal{A}, \mathcal{B})$ is the fraction of pairs with a strict inequality $s_a > s_b$. The target is $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C}) > 0.8$, where $\mathcal{P}$ is the five structured sites and $\mathcal{C}$ is the four clean sites.

At full rewrite, abrupt activation, and a median tail, over four seeds, that concordance equals 1 on both tasks, in every seed, for linear sites alone, for nonlinear sites alone, and for $\mathcal{P}$ against the noise site. On Income the linear score is 0.253 on the upper tail and 0.024 on the whole window, because hours worked pile up at 40 and the median tail is about 23% of the rows. The noise site has skill 0.008 and $\Delta^{\mathrm{cf}} = 0.001$, while its ungated contrast is $\tau^2 = 0.096$, larger than the clean contrast 0.015. The skill factor is what refuses to call that gap a driver. Pooled excess, without size matching and without the skill factor, ranks the Income noise site above every linear site: its concordance against the noise site is 0.400, which is exactly the two nonlinear sites beating the noise site and the three linear sites failing to. Label total variation and the covariate arms (domain AUC, RBF MMD) do not separate $\mathcal{P}$ from $\mathcal{C}$. The joint classifier on (X, Y) clears 0.8 against clean sites on average and ranks the noise site as the stronger departure.

A Pareto rank on $(\Delta^{\mathrm{cf}}, \tau^{2,\mathrm{g}}, \text{time jump})$ has the same concordance of 1. Mean ranks on Income are 1.875 (nonlinear), 3.250 (linear), 6.625 (clean), and 6.750 (noise). The undominated front is mostly the nonlinear sites. Linear sites are often dominated by a nonlinear site, because the nonlinear rule rewrites every row, and they remain strictly ahead of every clean site and of the noise site. The time curve of Δ^{cf}, on windows of four batches, leaves zero after $t^\star$ and levels off at batch 11, the first endpoint whose window lies entirely inside the drift. An online threshold at the maximum of the first two endpoints plus 0.01 hits every nonlinear site and, on Public Coverage, every linear site. Income linear sites hit in 0.833 of replicates. Income clean sites cross the threshold with rate 0.375; those crossings have mean absolute error 4.17 batches, so the typical alarm is not a call at $t^\star$. The derivative of the curve marks the start of the plateau, about one window later.

Concept-group logistic scores on (Z_1, Z_2), on $Z_1 Z_2$, and on $(\sin Z_1, Z_3^2)$ name the mechanism only on the region where it was planted. Top-five precision of Δ^{cf} is 1 in every seed. On the full window, product and sine-square recall are 1 and linear recall is 0.500. On the upper tail, linear recall is 0.917 and product recall falls to 0.375. Shift-permutation importance on a drifted site recovers the planted coordinates (AUC at least 0.906 on linear sites). Weighting those uploads by Δ^{cf} raises Income recovery AUC from 0.869 to 0.917. Leave-one-site-out influence on the uniform shift vector is about 0.01–0.03 and is not larger for the drivers than for the clean sites. The same negative result is Table 25. The object that separates drivers is Δ^{cf}, computed from the site's own labels, before the aggregate is formed.

The 0.8 target is not uniform in the design. On Income, it holds for rewrite probabilities 0.7 and 1 down to a linear tail of about 17% of rows, abrupt or gradual. It falls below 0.8 at rewrite probability 0.4 (mean concordance 0.783, worst seed 0.650, on the main tail), and at a tail of about 8% combined with rewrite probability 0.7 and a four-batch ramp (0.683, worst seed 0.500). On that weak main-tail slice the whole-window score, without the region maximum, has concordance 0.433.

Table 25: TabReD FSDS leave-one-site-out. Random is the deployment partition. Feature-rank is the sensitivity. Entries are mean±SD.

protocol	task	n	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	ρ_{unif}	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$
feature-rank	ALL	9	0.243 ± 0.241	0.057 ± 0.017	0.231 ± 0.237	0.047 ± 0.016	0.052 ± 0.029
feature-rank	cooking-time	3	0.300 ± 0.183	0.053 ± 0.008	0.299 ± 0.199	0.045 ± 0.015	0.036 ± 0.013
feature-rank	ecom-offers	3	0.264 ± 0.093	0.057 ± 0.006	0.245 ± 0.088	0.057 ± 0.005	0.061 ± 0.019
feature-rank	weather	3	0.165 ± 0.419	0.060 ± 0.031	0.148 ± 0.400	0.040 ± 0.024	0.060 ± 0.047
random	ALL	45	0.889 ± 0.106	0.054 ± 0.052	0.884 ± 0.115	0.049 ± 0.042	0.064 ± 0.048
random	cooking-time	15	0.807 ± 0.144	0.101 ± 0.068	0.794 ± 0.159	0.090 ± 0.049	0.114 ± 0.047
random	ecom-offers	15	0.921 ± 0.049	0.023 ± 0.013	0.923 ± 0.044	0.022 ± 0.012	0.030 ± 0.017
random	weather	15	0.939 ± 0.033	0.040 ± 0.014	0.934 ± 0.034	0.035 ± 0.011	0.048 ± 0.024

Table 26: Feature-rank cells: largest leave-one-site-out $|\Delta|$ by aggregator.

task	split	$\rho_{n\|S\|}$	$\|\Delta\|_{n\|S\|}$	$\|\Delta\|_{n}$	$\|\Delta\|_{\text{unif}}$	$\|\Delta\|_{\text{med}}$	$\|\Delta\|_{\text{rank}}$
cooking-time	num_184	0.092	0.054	0.047	0.061	0.027	0.031
cooking-time	num_38	0.437	0.045	0.054	0.042	0.031	0.048
cooking-time	num_123	0.373	0.060	0.062	0.032	0.051	0.053
ecom-offers	num_67	0.347	0.062	0.057	0.062	0.081	0.073
ecom-offers	num_104	0.283	0.059	0.067	0.053	0.042	0.056
ecom-offers	num_60	0.163	0.050	0.050	0.055	0.059	0.077
weather	num_5	−0.025	0.031	0.029	0.022	0.029	0.059
weather	num_84	−0.126	0.055	0.038	0.031	0.037	0.039
weather	num_98	0.645	0.093	0.068	0.067	0.114	0.054

1.6 Summary

Each site uploads one simplex vector, once. Four questions fix the procedure. When the weight on a site changes, and to what. How the aggregate responds when individual sites are corrupted. How a shift driver is justified across sites that need not share one concept. What the Matrix Bernstein bound supports. The four answers follow.

When and how to aggregate. Use $w_k = 1/K$ when the sites are noisy copies of one shift and have similar size. On TabReD the largest site fraction is 0.24 and the gain of $n|S|$ over the uniform average is about 0.01. Change the weight only as in Table 1. Under high size mismatch, set $w_k \propto n_k$ or use $n|S|$: Spearman with the mixed pool rises from 0.620 to 0.905, and the gain grows from +0.24 to +0.37 as heterogeneity goes from 0 to 1. When one or two vectors are replaced, use the coordinate-wise median or client-agreement Shapley. Two random replacements leave the uniform Spearman at 0.055 and the median at 0.741. When the variance estimates are bounded below and the wrapper is a covariate score, use inverse-variance weights: the WhyShift domain classifier moves from 0.848 on one fold to 0.918. When the wrapper is a concept score, set $w_k \propto \Delta_k^{\text{cf}}$. Sample-size weights lower WhyShift pseudo-outcome risk from 0.410 to 0.185, while Δ^{cf} weights raise Income shift-recovery AUC from 0.869 to 0.917. When every weight returns the same Spearman and that Spearman misses the whole batch, leave the weights alone. Cooking-time `num_105` is this case: pairwise concordance 0.884, site-to-whole correlation −0.198, and aggregator headroom about zero.

How robust the aggregate is. On the product grid of Table 11, precision at K can remain 1 after Spearman of the mean has fallen to 0.719, which is the cell with all twelve sites corrupted, full local support, and SNR = 0.5. The selection itself fails at SNR = 0 with twelve corrupted sites, where the mean AUC is 0.479 and precision is 0. One clean site preserves the mean (precision 0.92) and not the median (precision 0.14). Thin support weakens the median before any corruption: at $\alpha = 0.25$ and $n_{\text{bad}} = 0$, median precision is 0.29 against 1 for the mean. A corrupted site that reports variance ratio 0.44 takes inverse-variance mass 0.17, and Assumption (A6) is the floor that stops this. Uniform weights and the median keep Spearman 1 with the clean covariate aggregate under Dirichlet imbalance down to concentration 0.05. Five covariate-shift sites pooled with five concept-drift sites correlate about 0.30 with that clean aggregate. Five corrupted centers on SECOM, Diabetes130, TEP, and TCGA leave the selected set largely intact; at ten corrupted centers the mean-share AUC is 0.708, 0.920, 0.580, and 0.977 on those four sets. Equal FEMNIST blocks recover the pooled predictive

ranking at Spearman 0.996. Deleting one site moves a TabReD aggregate by about 0.05 whether the aggregate agrees with the whole batch or not.

How a primary driver is found. Name the driver before aggregating. The score Δ^{cf} in (5)–(6) is a size-matched Brier gap times local skill, maximized over the recent window and the two sides of the burn-in median of X_1. On ten sites that share the ACS empirical $P(X)$, with four clean sites, five structured concept shifts, and one Bernoulli(1/2) site, this concordance equals 1 at full rewrite and abrupt activation, on both tasks and in every seed, including against the noise site. The Income noise site has ungated contrast $\tau^2 = 0.096$ and $\Delta^{\mathrm{cf}} = 0.001$, because its skill is 0.008. Mean Pareto ranks on Income are 1.875 (nonlinear), 3.250 (linear), 6.625 (clean), and 6.750 (noise). The 0.8 target holds at rewrite probabilities 0.7 and 1 down to a linear tail of about 17% of rows. It is 0.783 at rewrite probability 0.4, and 0.683 on an 8% tail with a four-batch ramp at rewrite probability 0.7. Leave-one-site-out influence stays near 0.01–0.03 and does not separate the drivers.

Theoretical support. A simplex upload satisfies $\|V_k\|_2 \le 1$. For the uniform aggregate, (2) gives, with probability at least $1 - \delta$,

$$\left\|\bar{V} - \mathbb{E}\bar{V}\right\|_2 \le \sqrt{\frac{2\log(2p/\delta)}{K}} + \frac{2\log(2p/\delta)}{3K}.$$

Assumptions (A1)–(A4) carry the same bound to a weighted aggregate through K_{eff} and $R_\star$. The scalar bound (3) controls one contrast, and the coordinate-wise bound (4) controls all p coordinates, with dimension entering only as $\log(2p/\delta)$. Under the margin condition (A5), an ℓ_∞ deviation below half the margin makes the empirical Top-K set equal the population Top-K set. Inverse-variance weights additionally require the variance floor (A6); truncated or equal weights restore (4) when the floor fails. Sites are independent across k, coordinate dependence inside a site is unrestricted, and the $p \times p$ covariance is not estimated. Communication is $O(Kp)$ floats, one upload per site. The bound concentrates $\bar{V}$ around its expectation. The driver score decides which sites enter that expectation.

Streaming updates, modality-wise pseudo-outcome contributions, and a differentially private upload that preserves the same concentration remain open.

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